

Requirements

This document lists all software dependencies required to run the bug report classification tool.

Operating System

- macOS (Apple Silicon or Intel), Linux, or Windows 10+

Python Version

- Python 3.9 or higher (Python 3.11 recommended)

Required Python Libraries

Install all dependencies using the command below. All versions are pinned for reproducibility.

Library	Version	Purpose
pandas	≥ 2.0.0	Data loading and manipulation
numpy	≥ 1.24.0	Numerical operations
scikit-learn	≥ 1.3.0	TF-IDF, Naive Bayes, metrics
xgboost	≥ 2.0.0	XGBoost classifier
scipy	≥ 1.11.0	Wilcoxon statistical test
nltk	≥ 3.8.0	Stopword removal
matplotlib	≥ 3.7.0	Figures and plots
seaborn	≥ 0.12.0	Enhanced visualisation

Installation Command

```
pip install pandas numpy scikit-learn xgboost scipy nltk matplotlib seaborn
```

NLTK Data

The following NLTK datasets are downloaded automatically on first run:

- stopwords corpus
- punkt tokenizer

Dataset

The five CSV datasets (pytorch.csv, tensorflow.csv, keras.csv, incubator-mxnet.csv, caffe.csv) must be placed inside the **data/** subfolder (already placed for you but for guidance). These are available from the course repository at: <https://github.com/ideas-labo/ISE/tree/main/lab1>

Hardware

- Minimum: 4 GB RAM, any modern CPU
- Recommended: 8 GB RAM (Apple M2 or equivalent)
- GPU is not required.